```
num_candidates = 2000
num_repeats = 10

incumbents = torch.zeros((num_repeats, num_queries))

for trial in range(num_repeats):
    ... ❶

    for i in tqdm(range(num_queries)):
        ... ❷

        sobol = torch.quasirandom.SobolEngine(1, scramble=True)
        candidate_x = sobol.draw(num_candidates)
        candidate_x = bounds[0] + (bounds[1] - bounds[0]) * candidate_x

        policy = botorch.acquisition.max_value_entropy_search.qMaxValueEntropy(
            model, candidate_x
        )

        ... ❸
```

❶ Randomly generate the initial training data.

❷ Record incumbent value and retrain model.

❸ Find the point maximizing the acquisition score, query the objective function, and update the training data.

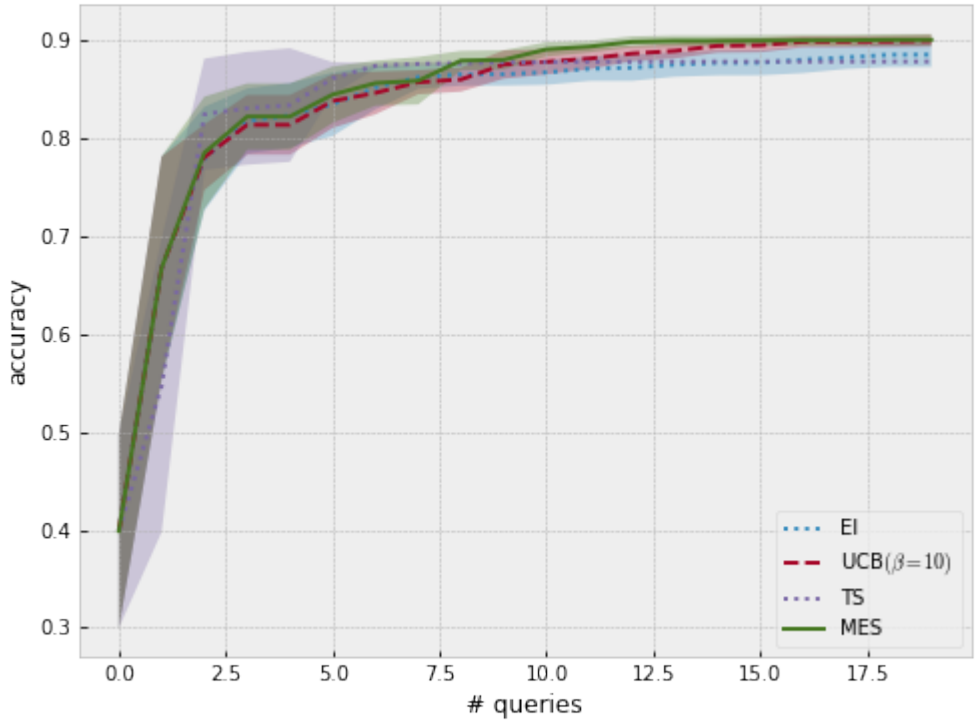


Figure A.15 Progress made by Max-value Entropy Search. The policy performs the best out of the fou policies shown.

Figure A.13 shows the optimization performance of MES. We see that the policy is competitive against all the BayesOpt policies that we have learned thus far.

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